

Question	Marking Guidance	Mark	Comments
10.1	1. Mutations/genotype/alleles; 2. Environment/habitat OR (Natural) selection; 3. Epigenetics; 4. Crossing over; 5. Independent segregation/assortment (of homologous chromosomes); 6. Random fusion of gametes OR Random fertilisation;	2 max (2 x AO2)	2. Accept named different habitats, e.g. 'walls' and 'trees/trunks'. 2. Accept selection pressure e.g. predation. 4. Accept recombination. 5. Accept 'Random assortment'.

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10.2	1. Provides camouflage; 2. (So) not seen by predators/prey OR Less predation OR Obtain/catch (more) prey;	2 (2 x AO2)	1. Accept description of camouflage. 2. Accept 'stops predation'. 2. Accept descriptions of reduced predation e.g. 'fewer are eaten'.

Question	Marking Guidance	Mark	Comments
10.3	Mark point 1 required for max marks 1. (Geckos in) same habitat/environment/area OR No geographical isolation/separation (between geckos); 2. (Possibly) allopatric speciation as different (areas of same) habitat(s)/environment/area OR (Possibly) geographical isolation/separation as different (areas of same) habitat(s); 3. (Could lead to) separate gene pools OR Reproductive isolation; 4. Mutation(s);	5 max (5 x AO3)	2. Accept 'walls' and 'trees/trunks' as different habitats. 3. Accept 'can't interbreed' in correct context, i.e., not when describing a species. 3. Accept 'no gene flow'. 4. Reject mutation(s) if context incorrect e.g., 'mutate to adapt', 'mutation caused by selection'. 5. Accept description of selection e.g. 'favoured'.

	5. Selection for (both) extremes/colours OR Disruptive selection (occurs) as two extremes/colours; 6. (Analysis shows that) diurnal geckos are a distinct (genetic) group; 7. (Genomes/DNA indicates geckos are) same species;		5. Accept selection against 'middle' 6. Accept 'nocturnal' for 'diurnal'.
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10.4	1. Compare <u>DNA</u> base/nucleotide sequence OR Compare banding/position of DNA fragments; 2. A distinct (group) will have different alleles/DNA/banding (from other group/s) OR If not distinct (group) will have similar alleles/DNA/banding (to other group/s) 3. DNA sequencing is automated/computerised OR Genetic/DNA fingerprinting is automated/computerised OR PCR amplifies DNA/genes OR Genetic fingerprinting/electrophoresis separates fragments/genes/alleles OR Use of DNA probes/hybridisation to identify genes/alleles;	3 (3 x AO2)	1. Idea of 'comparison' must be conveyed. 1. Accept alleles/VNTRs for 'DNA fragments'. 1 and 2. Accept genes for 'DNA fragments' in 1 but reject genes in mark point 2. 1 and 3. Ignore 'gene machine'. 2. Accept in context of either nocturnal or diurnal group being the distinct group. 2. Accept not 'closely related' for 'distinct (group)' and 'closely related' for 'not distinct (group)'. 2. Reject 'species' for 'group'.

Question	Marking Guidance	Mark	Comments
10.5	1. Marking not toxic so does not affect survival OR Marking not visible to predators OR	3 (1 x AO1, 2 x AO2)	1 and 2. Ignore births, deaths, reproduction, immigration, emigration. 1. Accept 'does not cause harm/death' for

	Marking does not wash/rub off so recaptured (geckos) identified; 2. Time/delay after release so (geckos) spread (in the population) OR Time/delay before recapture so (geckos) spread (in the population); 3. (Population =) (number in) first sample × (number in) second sample divided by (number) marked in second sample / number recaptured;		'does not affect survival'. 1. Idea of marking affecting visibility required in relation to predators. 2. Accept 'after marking' for 'after release' 3. Accept the correct equation/formula.
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